MODULE 8

Social Media and Web Intelligence

Certified Cyber Criminologist - Police / LEA Training

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What You Will Learn

The internet never forgets - but only a trained investigator knows where to look.

- **Web intelligence** - what it is and how its three pillars work
- **OCR and EXIF** - reading hidden text and metadata inside photos
- **The Wayback Machine** - recovering deleted websites as court evidence
- **Fake-profile detection** - the 10 red-flag checklist, scored

What You Will Learn (2)

And the discipline that keeps your evidence - and your career - safe.

- **Lawful monitoring** - exactly where your legal limit lies
- **Screenshots that survive** - URL, date-time, hash, every time
- **Chain-of-custody notes** - the 5-line habit of court-ready officers
- **5 common mistakes** - the traps that sink real cases

One Photo. One Board. One Arrest.

A fraudster posted a photo with his new car - "the fruit of hard work." In the background, a shop board was visible.

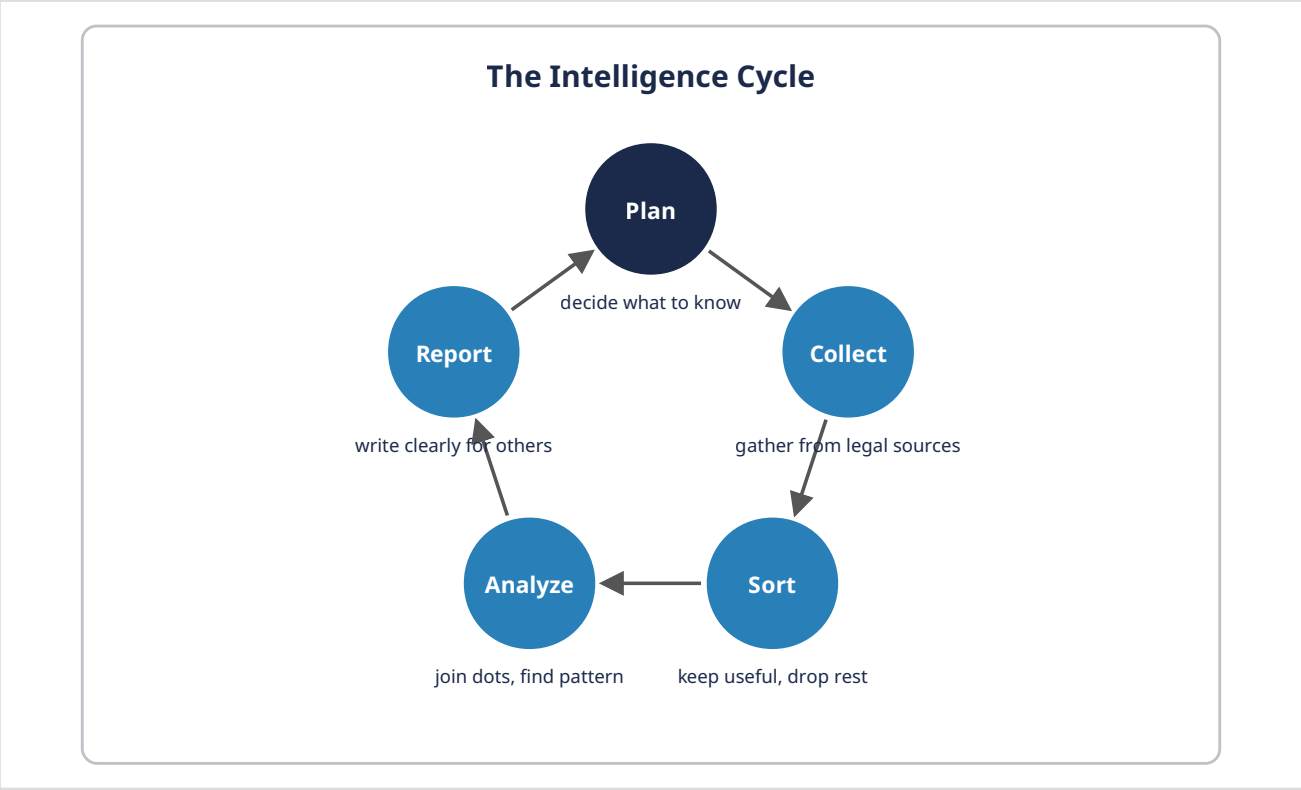
- The investigator **read the board**, searched it on Google, found the town
- Contacted that town's police - **arrest in three days**
- Lesson: every photo leaks something. Your job is to **see it**

What Is Web Intelligence?

Web intelligence means drawing conclusions from information already on the internet - news, blogs, forums, social media.

- **Three pillars:** search in the RIGHT PLACE, search in the RIGHT WAY, JOIN what you find
- An ordinary person **looks at** the internet. An investigator **searches** it.
- It runs in a cycle: plan, collect, sort, analyze, report - **never jump to conclusions**

The Intelligence Cycle



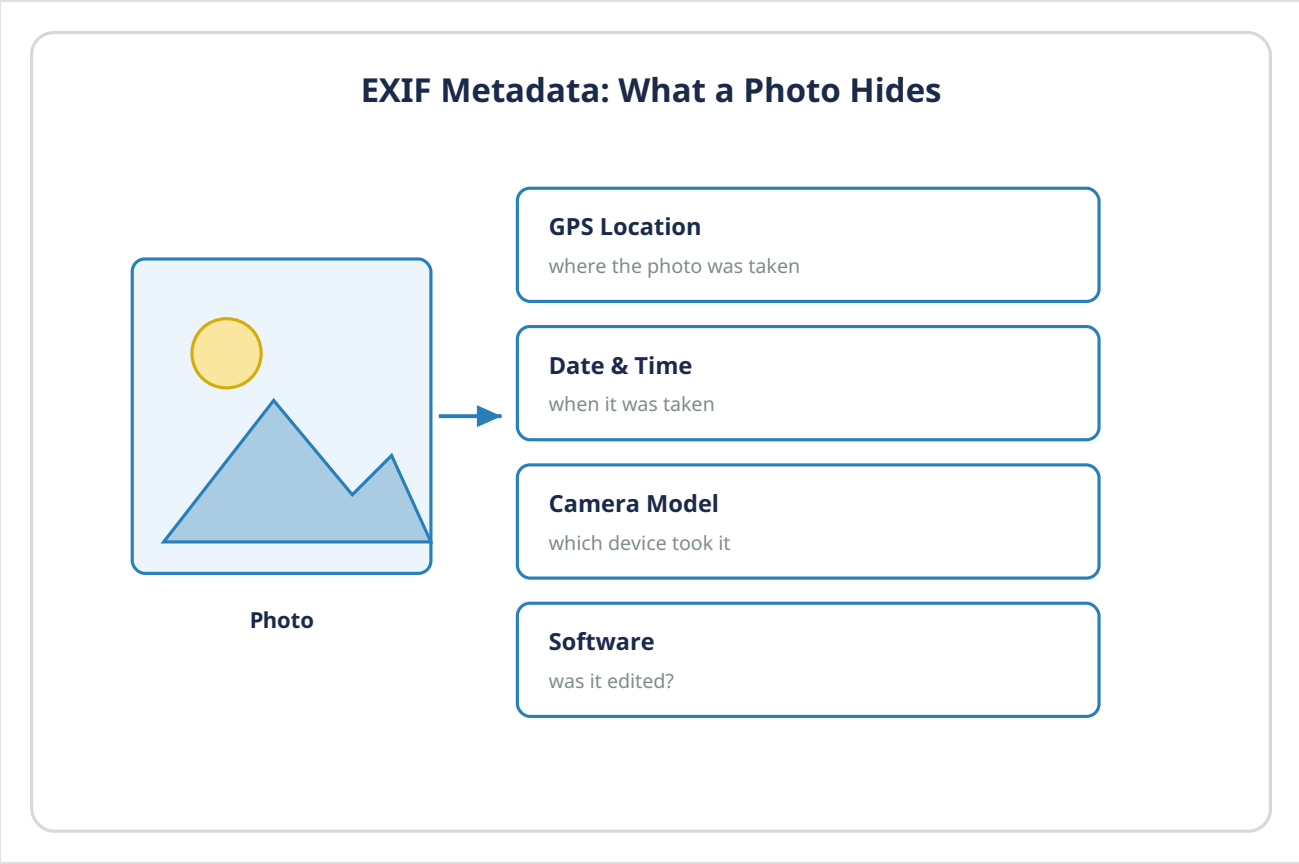
- **Plan first:** decide exactly what you need to know
- Collect, then sort useful from useless, then analyze
- Report in clear language so others can act on it

OCR: Words Hidden in a Picture

OCR - Optical Character Recognition - reads words hidden inside pictures. Google Lens does it free.

- Number plate blurred in a photo? OCR reads it - **owner traced via RTO record**
- Fake company's brochure photo? OCR extracts the phone number - **found on 3 more fake sites**
- Threatening screenshot? OCR pulls the visible time-date - **matched with call records**

EXIF Metadata: What the Photo Tells



- GPS coordinates, date-time, camera model - hidden inside every photo
- Criminals strip it (exiftool -all=); social media strips it too
- No metadata is ALSO information - it tells you they were careful

EXIF in Practice

ExifTool reads a photo's hidden metadata in seconds. Practice only on your own photos.

- **GPS Latitude/Longitude:** drop the coordinates into Google Maps - verify the location
- **Create Date vs Modify Date:** if they differ, the photo was edited after capture
- **WhatsApp test:** send yourself a photo, download it, re-check - see what WhatsApp removed

The Wayback Machine

Criminals delete their websites before fleeing. web.archive.org remembers them anyway.

- **Save Page Now** the moment you see a fake site - do not delay a day
- Old snapshots reveal promises, phone numbers, bank details - **all court evidence**
- Snapshot the WHOLE site, not just the homepage - then hash the saved page

Search Commands That Find Everything

Four small commands. Enormous investigative power - on public information only.

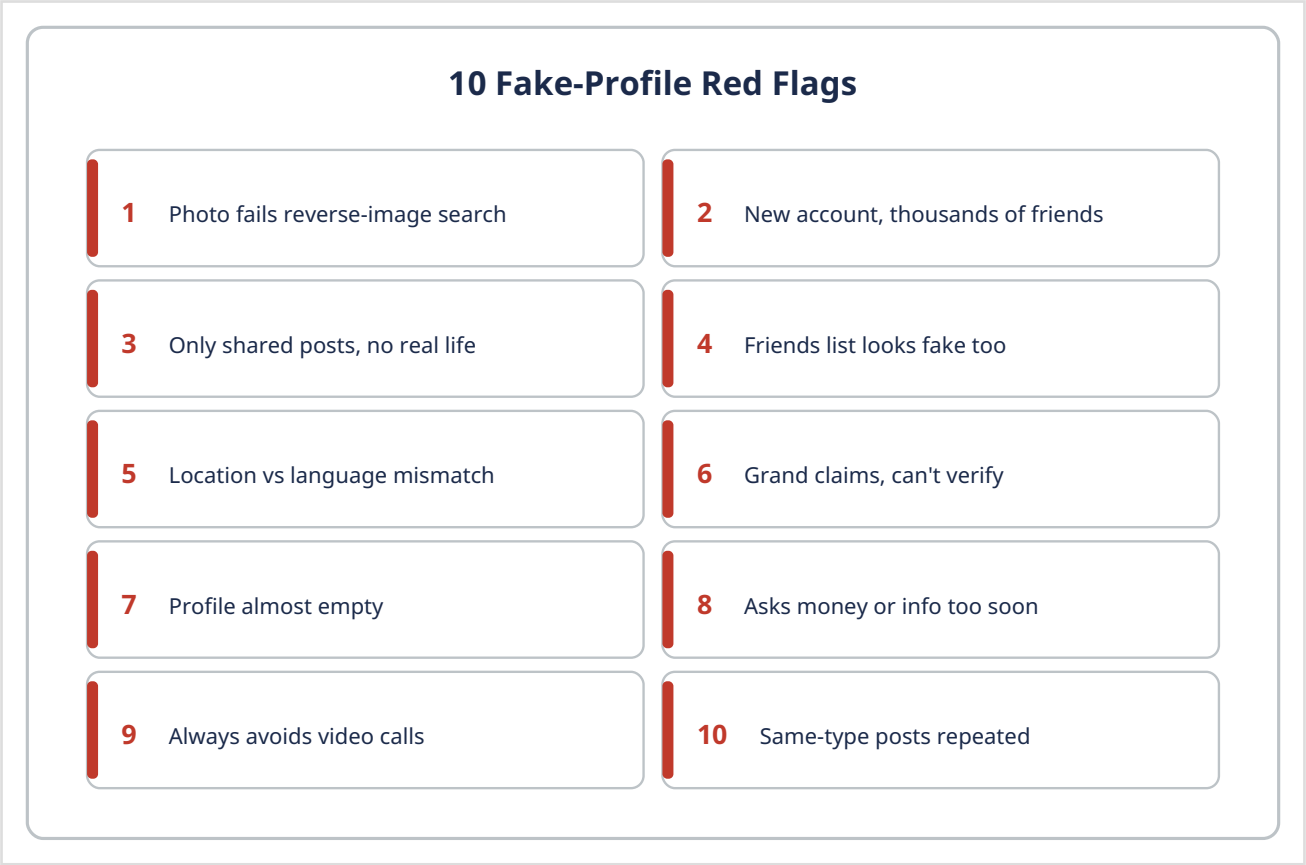
Command	Use	Example
site:	Search inside one site	site:linkedin.com "cyber security" india
filetype:	Find one file type	filetype:pdf "company name"
intext:	Word inside the page	intext:"98200 12345"
"quotes"	Exact phrase	"98200 12345" - every public mention

Scraping: The Legal Workflow

Scraping collects website data automatically - but every site has rules, and breaking them is wrong.

- **Step 1:** read the site's rules - is automated access permitted?
- **Step 2:** take only what the case needs; note every page's URL and date
- **Step 3:** keep data in the case file - **never publish it**
- Rule: what's public, see it. What's hidden, get it through law.

10 Red Flags of a Fake Profile



- Photo, account age, post pattern - check in that order, never judge by photo alone
- Score it: 0-2 low suspicion, 3-5 caution, **6+ very suspicious**
- High score = suspicion, NOT evidence - legal process starts after

The Four Flags That Catch Most Fakes

Ten red flags, one checklist. These four catch the majority of fake profiles:

- **Reverse-search the photo** - a model's photo under another name means fake
- **Account made yesterday, 5,000 friends?** The math doesn't add up
- **Only shared posts, no real life** - real people post their lives
- **Says Delhi, writes American English** - catch the contradiction

Reading a Profile: The Right Order

Read a suspect profile in THIS order - haste makes you miss red flags:

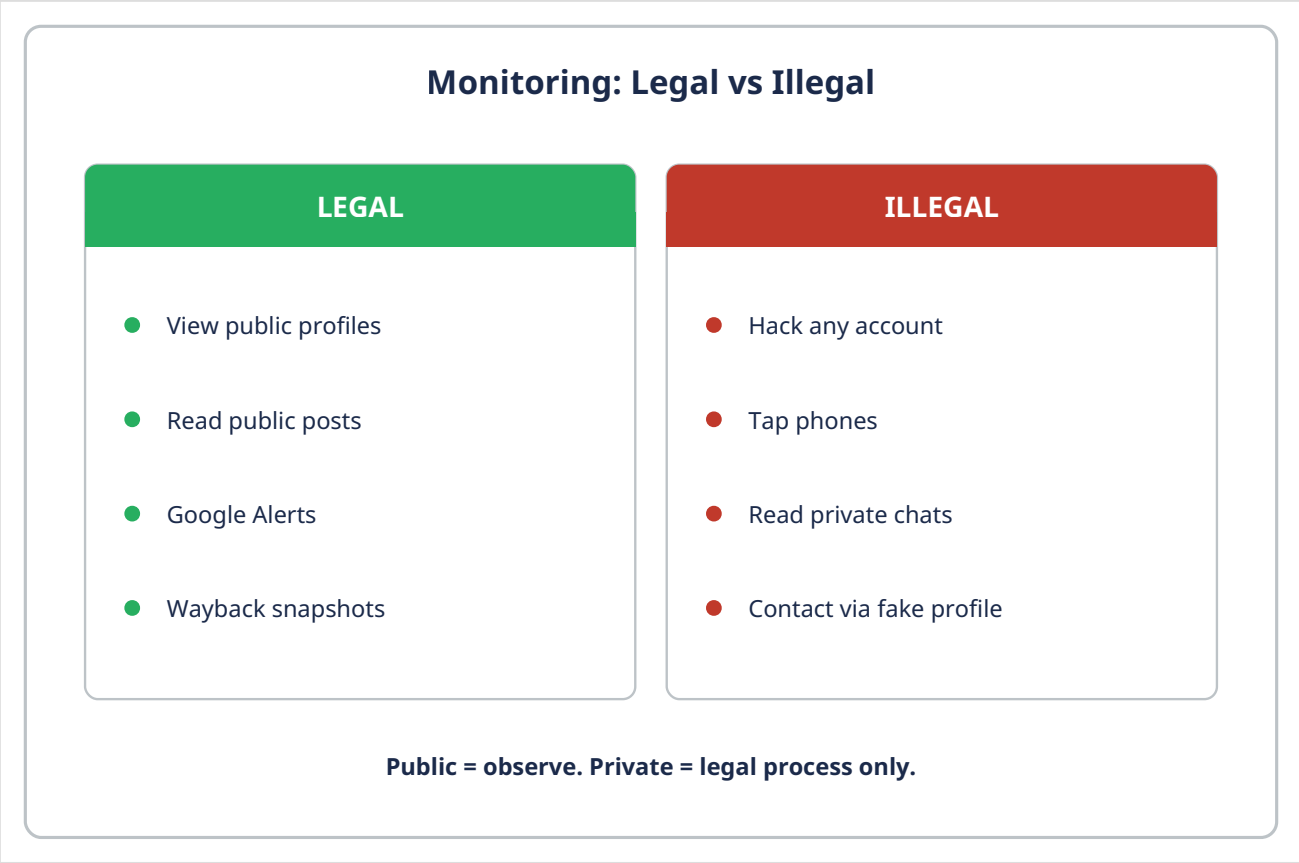
- **1. Photo** (reverse search) → **2. Account age** vs oldest post → **3. Post pattern**
- **4. Friends list** (fake webs follow each other) → **5. Language style**
- **Video-call test:** a fake profile will ALWAYS avoid it
- Never decide after the photo alone

Telegram Intelligence

Crime is growing on Telegram. Watching public channels is legal - joining them is not.

- **Public channels** selling leaked data: view and note, never buy anything
- Bot names, group member lists, forward paths - **all reveal the network**
- Report properly: channel link + 2-3 clear screenshots on cybercrime.gov.in
- A complaint without evidence gets slow action

Lawful vs Illegal Monitoring



- **Legal:** public posts, public profiles, news searches - anyone can do this
- **Illegal:** hacking accounts, tapping phones, reading private chats
- Private data needs legal process - a senior's verbal order is NOT enough

Know Your Legal Limit

Monitoring is watching, not entering. The line is bright:

- Public post viewing needs **no permission**. Private accounts, CDR, location need court or competent authority
- As a trainee, your scope is **PUBLIC information only**
- Demand a **WRITTEN order** before touching private data - it protects you
- Curiosity beyond scope is not investigation - it's a liability

The Weekly Monitoring Log

Professional monitoring runs for months. One page per week, six fields:

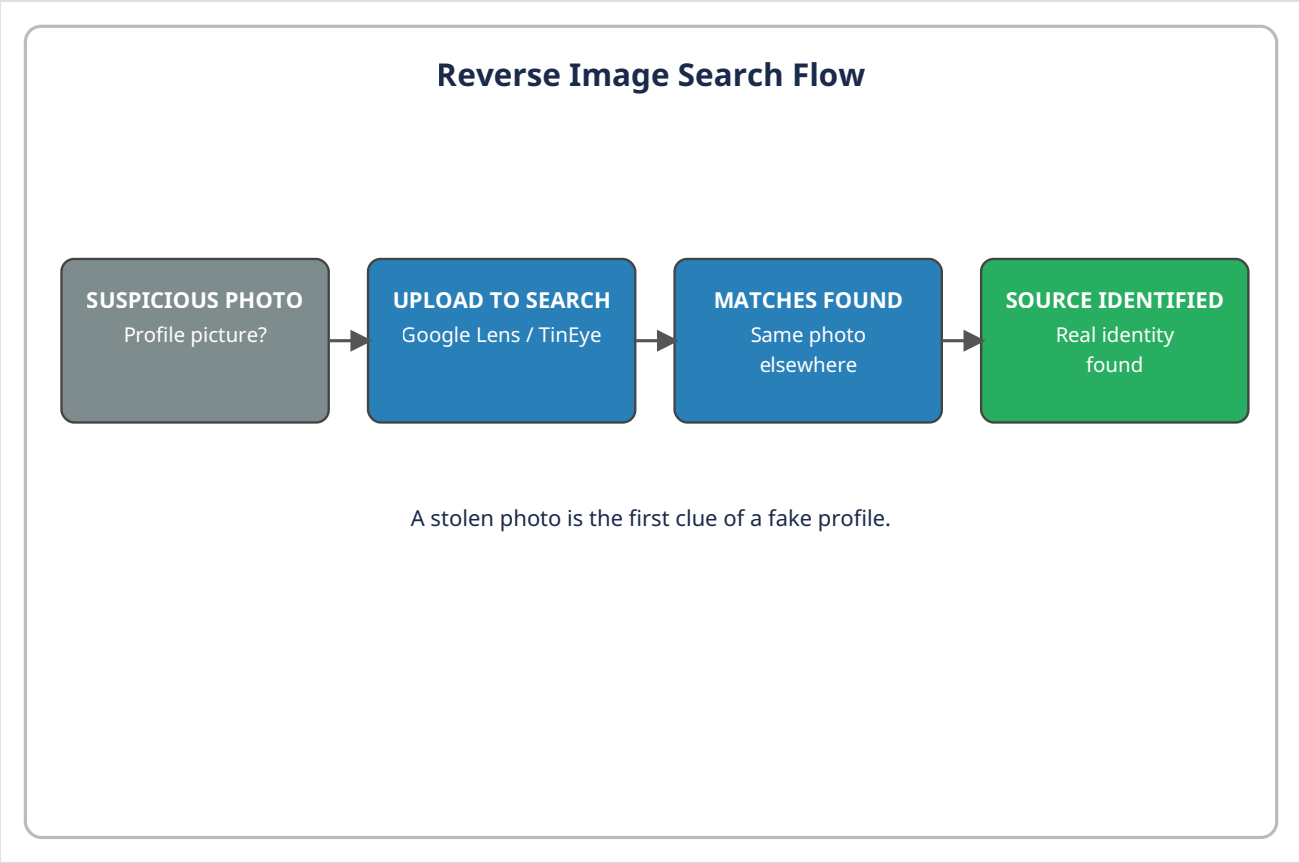
- Week's dates + subject under watch (suspect, company, keyword)
- Sources checked: Google News, profiles, forums, Wayback Machine
- New information found - or write **"no change"** (that IS information)
- Signs of change + what to check next week

Signs the Suspect Sensed Danger

Watch for these - they mean the suspect knows someone is watching:

- Profile deleted, name changed, photo removed **overnight**
- Old posts vanishing = destroying evidence - **snapshot NOW**
- Sudden silence after months of activity
- Log every change with its date - the timeline builds itself

Making a Screenshot Evidence



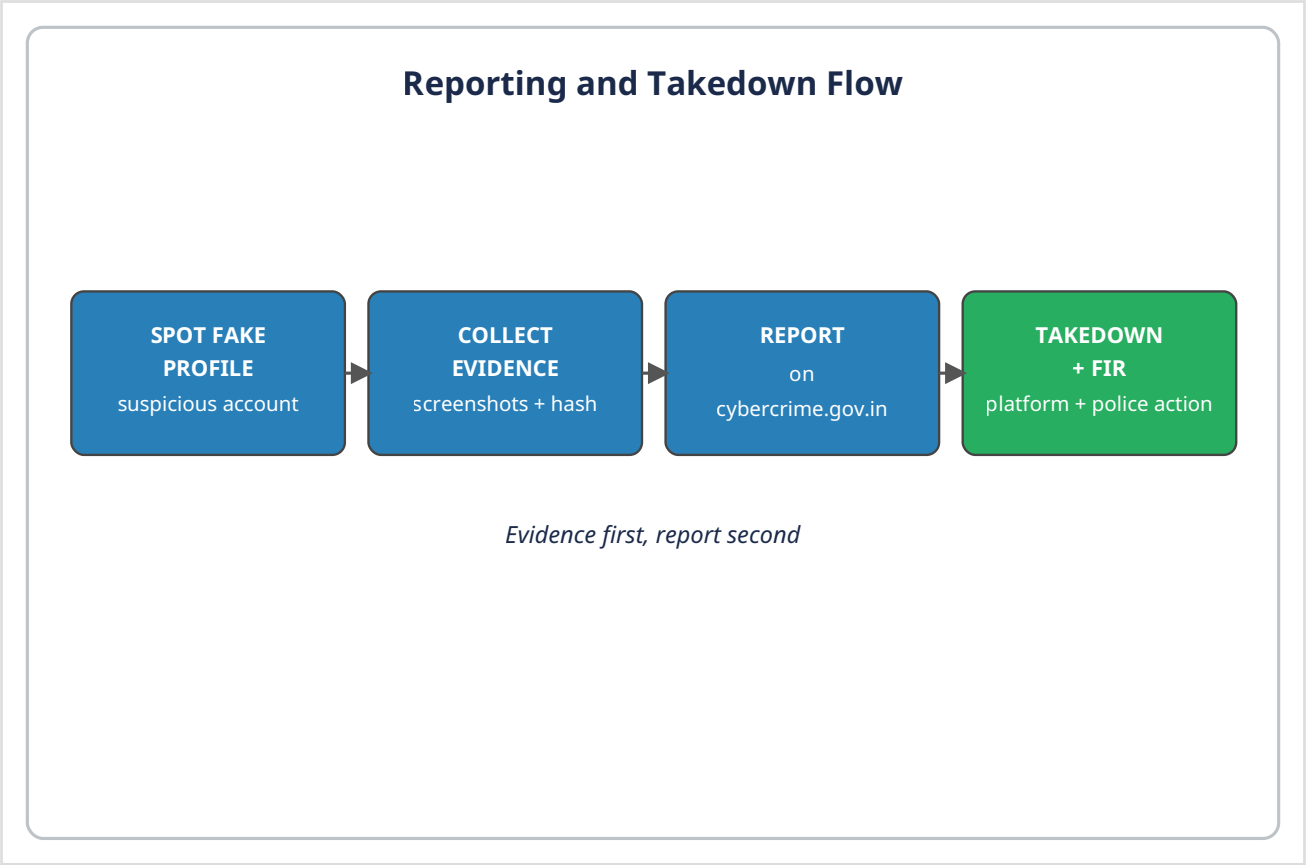
- Full URL + date-time + one-line note + **SHA-256 hash** - all four, every time
- Never crop, never edit - an edited screenshot stands weak in court
- Long pages: use full-page capture - half a page raises questions

The Chain-of-Custody Note

Every archived page, photo or screenshot gets a 5-line note. Without it, the hash means nothing:

- Date-time, **complete source URL**, what you did (snapshot / hash)
- SHA-256 hash of the file + your name and designation
- Remark: "page was public, no login required"
- Copy this format into your notebook - use it in EVERY lab

Reporting and Takedown Flow



- Note channel name, username, link + 2-3 clear screenshots
- File on cybercrime.gov.in with evidence attached
- Follow up - documented complaints move faster

5 Common Investigation Mistakes

Cases are lost to these five mistakes more often than to clever criminals:

- **Trusting a single source** - always verify with two independent ones
- **Private curiosity** - the suspect's relatives' profiles are NOT investigation
- **No hash, no date** - months later you can prove nothing
- **Contacting the suspect from a fake profile** - damages YOUR case

The Fake-Profile Trap

The most tempting mistake - messaging the suspect from a fake account:

- It alerts the suspect and hands the defence a weapon against YOUR case
- If a senior asks you to do it, **ask for the written order first**
- Clean procedure protects both your safety and the conviction

Real Case: The Deleted Investment Site

A fake investment company defrauded crores of rupees - then deleted its website before fleeing:

- Investigators pulled old snapshots from **web.archive.org**
- Promises, phone numbers, bank details - all present in the archive
- Those old pages became **court evidence** and led to the accused
- Source The Economic Times

Real Case: Lessons in 5 Points

What the deleted-site case teaches every investigator:

- **Archive the moment you see it** - do not wait for permission to preserve
- Snapshot EVERY page of the site, not just the homepage
- Separately note phone numbers and bank details visible on the site
- The snapshot's date proves WHEN the information existed - guard it

Your Turn: Labs in This Module

Four hands-on labs - all legal, all on your own machine:

- **Lab 1:** reverse image search - verify whether a photo is original or stolen
- **Lab 2:** ExifTool - extract GPS, camera and dates from your own photos
- **Lab 3:** archive a webpage + SHA-256 hash - write a chain-of-custody note
- **Lab 4:** score 3 sample profiles on the 10-flag checklist

Key Takeaways

If you remember nothing else from Module 8, remember this:

- Everything on the web is a lead - **if you know where to look**
- Reverse-search the photo, read the metadata, archive the page
- **Hash everything. Date everything.** Verify with two sources.
- Public watching is legal - private entry needs legal process
- Suspicion is not evidence - the checklist decides direction, not verdict
- Monitor within the law, document everything

Review Questions

- 1. A suspect's photo carries GPS metadata showing his home. Can you raid tonight? List the legal steps first.
- 2. A profile scores 7/10 on red flags. A colleague wants to declare it fake publicly. Give three reasons you refuse.
- 3. Why does a cropped, edited screenshot fail in court even when it shows the truth?
- 4. Your senior verbally orders you to pull someone's private chats. What do you say - and why?
- 5. Design a 4-week monitoring plan for a suspected fake investment site. What do you log each week?

NEXT: MODULE 9

Dark Net and Deep Web Investigation

Tor, onion routing & investigator limits

Cyber Crime Investigator - Training Manual